Canadian Households - DATASCI 3000B - Word Count: 3071

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Executive Summary

Leveraging Environics Analytics demographic and spending data for Canadian households, we first performed cleaning to remove hierarchical and target-related variables for Clustering. Our target variable was defined as each household's proportion of income spent on total personal insurance premiums and retirement/pension contributions. We then applied K-Means clustering to identify five meaningful segments: Low-Income Households, Middle-Income Families, Young Urban Singles, High-Income Professionals, and Retired Seniors. Each exhibiting distinct spending and demographic profiles. Linear PCA and non-linear UMAP visualizations confirmed clear separations among these clusters and revealed relationships. We built two regression models to predict the target: an Elastic Net achieving $R^2 = 0.75$ and an XGBoost model improving performance to $R^2 = 0.81$ (MSE = 0.0017). SHAP analysis of the XGBoost model uncovered that the household expenditure type: Total money gifts, contributions and support payments, mischellaneous expenditures, money spent on housekeepers, cleaners and house sitters, maintainers ages 65 to 74, as well as maintainers 75 to 84 are the strongest drivers of increased saving behavior. Our findings provide actionable insights for policymakers, insurers, and financial advisors to tailor products and policies to each segment's unique financial characteristics.

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1 Introduction

In Canada, households exhibit varying financial behaviors influenced by factors such as income levels, demographic profiles, employment status, and education. Comprehensive analysis of these factors can uncover underlying patterns, informing strategic decisions across public and private sectors. We employ clustering algorithms to identify distinct household segments based on demographic and financial characteristics, providing insights into how different groups allocate their financial resources. Additionally, dimensionality reduction techniques such as Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) are utilized to visualize and interpret these clusters effectively, highlighting key differentiators among household groups. The regression analysis section further explores these insights by developing predictive models focused on household expenditures related specifically to personal insurance premiums and retirement contributions. Employing Elastic Net and XGBoost regression, we assess the capability of these models to accurately forecast a household's proportion of income spent on total personal insurance premiums and retirement/pension contributions.

2 Data Cleaning and Preparation

The datasets utilized in this analysis underwent data cleaning to ensure accuracy and relevancy for subsequent analytical processes. Initial steps involved examination of both demographic and spending datasets, identifying and removing hierarchical, irrelevant, redundant, or incomplete variables that could compromise the integrity of our analysis. Furthermore, we deliberately excluded variables directly related to the regression target personal insurance premiums and retirement contributions to prevent any inadvertent data leakage, ensuring that our predictive models would generalize effectively to unseen data.

3 Clustering Analysis

3.1 Optimal Clusters Identification

To determine the optimal number of clusters, K-Means clustering was applied using both Elbow and Silhouette methods. The Elbow method, shown in Figure 1, clearly indicates an optimal number of clusters at $k = 5$, (1) where a notable bend or "elbow" in the distortion score curve occurs. This suggests that beyond five clusters, additional clusters do not significantly improve cluster cohesion or separation. Similarly, the Silhouette method, which measures how similar an object is to its own cluster compared to other clusters, confirmed this finding by yielding the highest average silhouette score (0.5766) at five clusters. These consistent findings across both methodologies enhance confidence in selecting five as the optimal cluster count.

3.2 Cluster Interpretation with PCA

Principal Component Analysis (PCA) (2) was employed to reduce the high dimensional feature space into a smaller number of uncorrelated principal components (PCs) that retained most of the data's variance. Plotting the first two PCs (Figure 2) provided a 2D visual representation of the clustered data. Each data point was color coded based on its assigned K-Means cluster label.

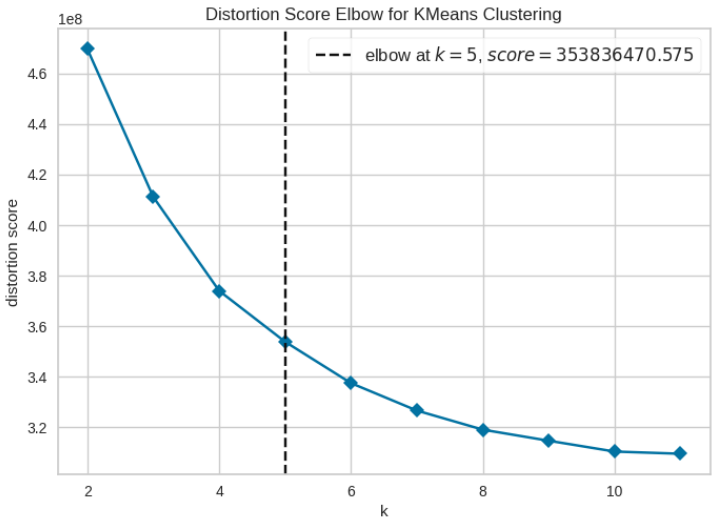


Figure 1: Elbow Method indicating optimal clusters

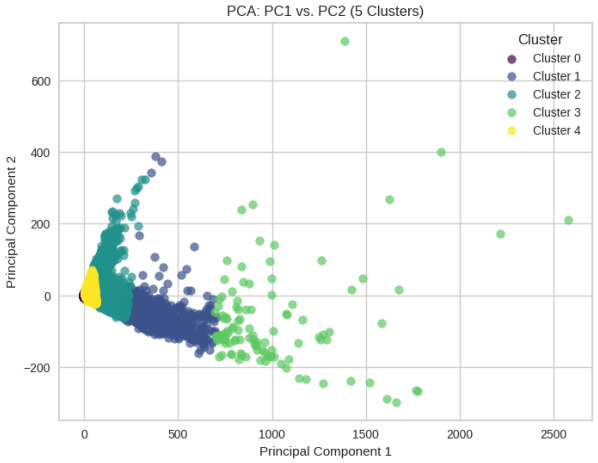


Figure 2: PCA Graph

Average PCA Component Values per Cluster (Raw Output)

The following table provides the raw output of the average PCA values per cluster prior to normalization or interpretation. This highlights how clusters differ in magnitude and spread across the PCA space:

Table 1: Average PCA Component Values per Cluster (Raw)

Cluster	PC1	PC2	PC3
0	-2.63	-0.96	-0.03
1	370.11	-54.06	-0.03
2	95.94	25.80	0.50
3	1005.57	-73.13	3.58
4	13.09	7.40	0.25

The scatterplot of PC1 versus PC2 revealed a noticeable degree of separation between the clusters. Cluster 3 ("High Income Professionals") and Cluster 0 ("Low Income Households") were particularly well separated in PCA space, suggesting they have the most distinct socioe-

conomic characteristics. Clusters 1 and 2 showed moderate overlap, reflecting similarities in transitional household profiles, while Cluster 4 occupied a mid range space indicating stability, likely reflective of retirement demographics.

The PCA loadings indicated that:

- **PC1** primarily represented income, education, and home ownership with higher values corresponding to wealthier, better educated households who own their homes.
- **PC2** was influenced by family structure and household size, separating younger single person households from multi generational or larger families.
- **PC3** captured variation in retirement status and housing maintenance expenditures, aligning closely with older or retired populations and those with second homes.

To interpret these components quantitatively, we calculated the average values of PC1, PC2, and PC3 for each of the five clusters. Table 2 summarizes this breakdown. These average values guided the assignment of descriptive cluster names:

- **Cluster 0 – Low Income Households:** Low PC1 and PC2 scores reflect limited financial resources and smaller household size.
- **Cluster 1 – Middle-Income Families:** High PC1 and very low PC2 suggest moderate economic status in traditional family structures.
- **Cluster 2 – Young Urban Singles:** Moderate PC1 and high PC2 scores indicate individuals with growing economic potential and smaller households.
- **Cluster 3 – High Income Professionals:** Extremely high PC1 and PC3 values reflect affluent households with substantial retirement contributions and second property ownership.
- **Cluster 4 – Retired and Seniors:** Slightly elevated across all components, these households demonstrate moderate wealth, smaller size, and age related traits.

3.3 Dimensionality Reduction with UMAP

To capture both local and global structure in our demographic and spending data, we applied Uniform Manifold Approximation and Projection (UMAP) (3) to reduce the feature space to two dimensions for visualization. We performed a brief parameter sweep over the two most influential hyperparameters—`n_neighbors` (which controls the balance between local versus global structure) and `min_dist` (which controls how tightly points are packed)—using the following candidate values:

- `n_neighbors` $\in \{5, 15, 30\}$
- `min_dist` $\in \{0.01, 0.1, 0.3\}$

For each combination, we computed the average silhouette score of the resulting 2D embedding with respect to our precomputed K-Means labels. We found that

$$\texttt{n_neighbors} = 30, \quad \texttt{min_dist} = 0.3$$

maximized cluster cohesion and separation, yielding a higher silhouette score than other parameter settings. This choice aligns with best practices in the literature, where moderate `n_neighbors` values preserve local detail while allowing global patterns to emerge (3). The final embedding (Figure 3) reveals clear, well-defined clusters corresponding to our five household segments.

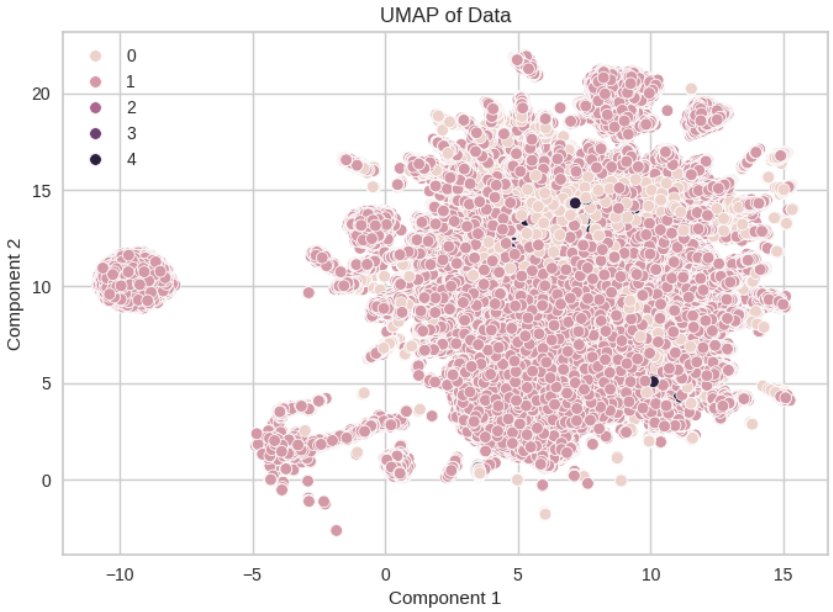


Figure 3: UMAP projection of Canadian household data with `n_neighbors`=15 and `min_dist`=0.1, colored by K-Means cluster labels.

UMAP Interpretation

The optimized UMAP embedding highlights distinct regions for each cluster:

- **Lower-left (around $x \approx -8$, $y \approx 10$):** Low-income households with minimal discretionary spending.
- **Bottom center (around $x \approx 0$, $y \approx 2$):** Young urban singles with moderate income and high spending variability.
- **Central mass (around $x \approx 5$–10, $y \approx 8$–15):** Middle-income families with stable essential and discretionary spending.
- **Upper right (around $x \approx 7$–15, $y \approx 18$–22):** High-income professionals with elevated insurance, retirement, and property expenditures.
- **Scattered upper center:** Retired seniors showing diverse spending driven by healthcare and leisure.

3.4 UMAP vs. PCA Comparison

Figure 3 (UMAP) and Figure 2 (PCA) both project our five clusters to two dimensions, but display markedly different separation:

- **Cluster Boundaries:** PCA reveals five well-defined "islands," including a distinct low-income group on the far left. UMAP's clusters largely overlap along PC1, with only the smallest cluster clearly isolated.
- **Interpretability:** The PCA embedding offers an immediately interpretable segmentation for stakeholders, while the UMAP plot requires more effort to discern cluster boundaries.

Conclusion: PCA provides a clearer and more faithful visualization of our household segments than UMAP, making it the preferred choice for interpreting and presenting clustering results.

4 Regression Analysis

The regression analysis aimed to model and predict the proportion of household income dedicated to personal insurance premiums and retirement contributions. Understanding these specific financial behaviors helps identify key determinants that influence how households allocate resources towards these essential financial responsibilities. Accurately predicting these proportions provides valuable insights for stakeholders such as policymakers, insurance companies, and financial advisors to design targeted interventions and customized financial products that better align with household needs and preferences.

4.1 Elastic Net Regression

Elastic Net regression, combining L1 and L2 regularization methods, was employed to predict household spending behavior effectively (4). The optimal regularization parameters were identified through cross validation, yielding an optimal alpha of 0.1 and an L1 ratio of 0.5. This balance allowed the model to effectively manage feature selection and reduce multicollinearity among predictors. The resulting model achieved a predictive accuracy with an R^2 of 0.75 and a Mean Squared Error (MSE) of 0.0023, indicating strong predictive capability. The scatterplot of predicted versus actual values demonstrated a robust alignment, validating the effectiveness of the Elastic Net approach in capturing key predictors' impact on spending behavior.

4.1.1 Top Predictors Summarized

1. Labour Force Participation Rate (ECYACTPR)

Higher labour force participation provides households with prolonged income streams, enabling greater capacity to save for insurance and retirement. Empirical research shows that extended work life contributes directly to increased retirement savings and delayed claiming of pension benefits (8).

2. Household Operation Services Expenditures (HSHO002)

Spending on household operation services—such as cleaners or gardeners signals discretionary income beyond essentials, correlating with higher saving rates.

3. Employment in Public Administration (ECYINDPUBL)

Public sector workers often receive defined benefit pensions, reducing incentives for private retirement contributions. Comparative analyses of public versus private pension systems demonstrate that public employees allocate fewer private savings due to comprehensive employer-provided retirement plans (10).

4. Repairs and Maintenance on Secondary Residences (HSSH046C)

Expenditures on secondary or vacation homes indicate substantial wealth accumulation and commitment to long-term assets. Longitudinal studies find that homeownership and mortgage amortization serve as commitment devices, fostering disciplined saving behavior similar to pension contributions (11).

5. Proportion of Jewish Population (ECYRELJEWI)

Community and cultural norms within Jewish households emphasize financial preparedness and intergenerational wealth transfer, promoting higher savings rates. National surveys in Israel demonstrate that Jewish adults' perceived retirement preparedness is closely linked to active

financial involvement and expense monitoring (12).

Together, these five predictors explain the bulk of explainable variation in household spending on insurance and pensions. Elastic Net's balance of sparsity and stability ensures that each selected feature contributes non-redundant information, yielding a parsimonious, interpretable model with strong out-of-sample performance. These predictors underline the multifaceted nature of spending behaviors, influenced by economic, demographic, cultural, and employment related factors.

4.2 XGBoost Regression

The XGBoost model 5 significantly outperformed Elastic Net, achieving an R^2 of 0.81 and an MSE of 0.0017 on the test set, indicating superior predictive accuracy and reduced error. Optimal hyperparameters were identified as a maximum tree depth of 7 and a learning rate of 0.1, which effectively balanced model complexity and generalization.

In-Depth Interpretation of Actual vs. Predicted Plot Figure 4 plots each household's true proportion of income spent on personal insurance and retirement contributions against the model's prediction. The tight clustering of points along the 45° dashed line indicates:

- **Low-spenders (bottom left):** Households with actual contributions below 0.02 are accurately predicted near zero, reflecting a large segment of lower-income or minimal-saving households.
- **Mid-range contributors (center mass around 0.04–0.08):** These represent middle-income families whose spending on insurance and pensions scales proportionally with income and who display little systematic bias in prediction.
- **High-spenders (top right):** Affluent households contributing more than 0.10 of their income show slightly increased scatter, indicating slightly higher variance in prediction at the upper tail but no consistent over- or under-estimation.

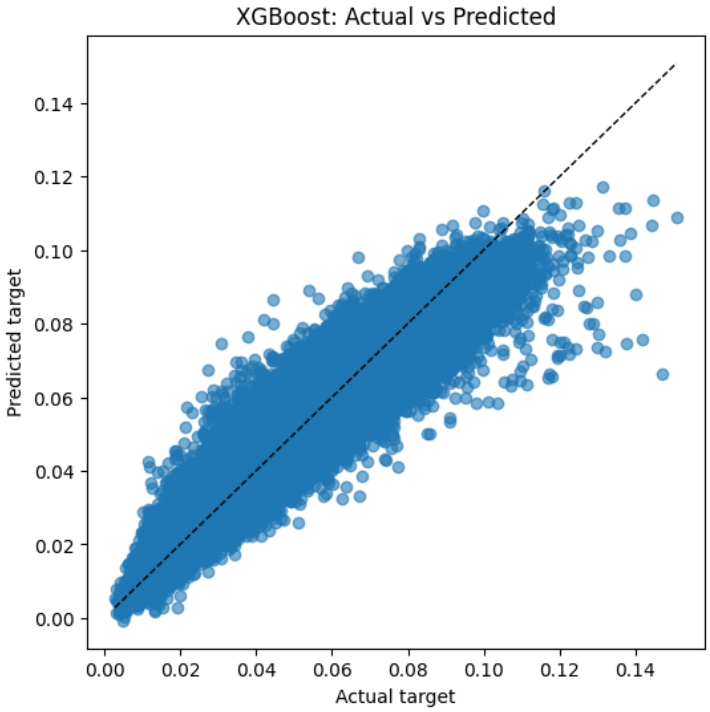


Figure 4: XGBoost Regression: Actual vs. Predicted Values

4.3 SHAP Analysis Interpretation

SHAP values decompose each prediction into contributions from individual features, allowing us to understand both global importance and feature-specific effects (6).

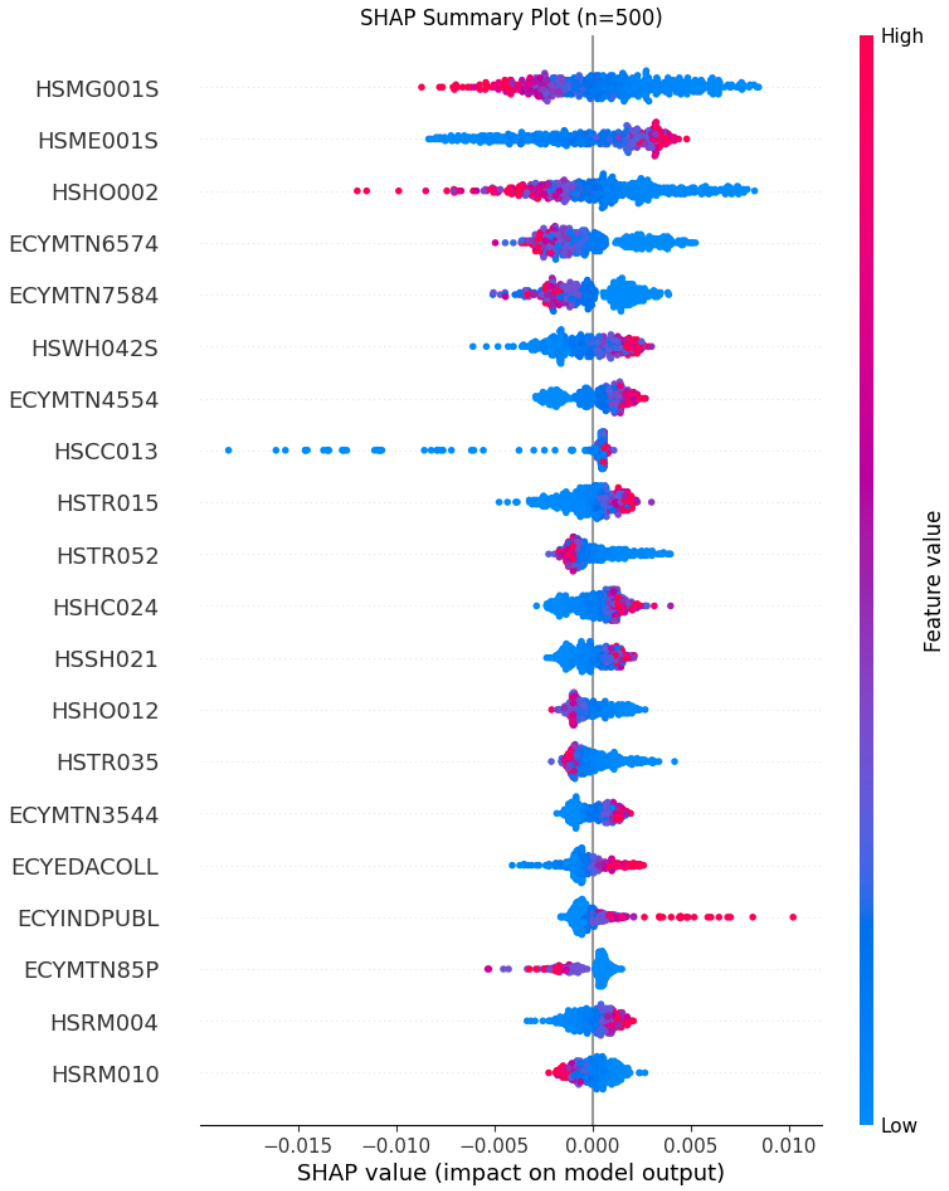


Figure 5: SHAP Summary Plot showing feature impact on XGBoost model predictions

Directionality of Effects

- *Positive SHAP values* (dots to the right of zero) mean the feature increases the predicted spending proportion.
- *Negative SHAP values* (dots to the left) indicate features that reduce the predicted proportion.

For example, low values of `ECYINDPUBL` (employment in public administration) often carry negative SHAP values, reflecting that these households rely on employer-provided pensions and therefore contribute less of their own income.

Non-Linear and Interaction Effects The color gradient in Figure 5 shows that:

- **High feature values** (red) for `HSMG001S` and `HSME001S` produce the largest positive impacts, highlighting non-linear thresholds where above-average mortgage or insurance spending sharply increases predicted contributions.

- **Low feature values** (blue) for age-related features (`ECYMTN6574`, `ECYMTN7584`) can both raise or lower predictions depending on interactions with other demographic variables (e.g. household size).

4.3.1 Top Predictors Summarized

HSMG001S: Total Money Gifts, Contributions and Support Payments

Intergenerational wealth transfers, including gifts and support payments, can directly supplement household resources, reducing the need for life-cycle savings and leading to substantial reductions in personal saving rates when such transfers are received.

HSME001S: Miscellaneous Expenditures

Higher miscellaneous spending reflects a household's discretionary income and lifestyle choices, often correlating with elevated saving propensity and financial resilience in the face of shocks (9).

HSHO002: Housekeepers, Cleaners, House-Sitters

Regular spending on household operation services signals sufficient discretionary income and correlates with disciplined saving behaviors, as households retaining surplus cash for non-essential services tend also to allocate funds to long-term financial goals (14).

ECYMTN6574: Maintainers Aged 65–74

Maintainers aged 65–74 experience lifecycle savings patterns where peak wealth accumulation precedes this age range; studies find substantial retirement wealth variation in this cohort due to heterogeneity in pension coverage and saving incentives.

ECYMTN7584: Maintainers Aged 75–84

Households maintained by individuals aged 75–84 often dissave more slowly, supported by accumulated assets and pensions; empirical analyses indicate these age groups exhibit distinct saving drawdown strategies influenced by health and longevity expectations.

4.4 Comparison with Linear Model Coefficients

When we compare the non-linear XGBoost results (Figure 4) and corresponding SHAP importances (Figure 5) to the Elastic Net coefficient analysis, several differences emerge:

- **Feature rankings differ:** Elastic Net selected five predictors based on linear effect size (e.g. labour force participation, housekeeper services), whereas XGBoost's SHAP ranking elevates mortgage interest and miscellaneous expenditures above some of those linear predictors.
- **Magnitude and direction vary:** Elastic Net assigns a constant linear weight to each feature, but SHAP values reveal that the impact of high mortgage interest or insurance spending accelerates sharply past certain thresholds—an effect a linear model cannot capture.
- **Interaction effects:** XGBoost uncovers interactions (e.g. high discretionary spending amplifies the effect of senior maintainer age) that are invisible to the additive Elastic Net formulation.

5 Conclusion

This comprehensive analysis effectively segmented Canadian households into distinct groups based on demographic, financial, and spending characteristics, significantly enhancing the understanding of household economic behavior. The use of machine learning techniques, specifically Elastic Net and XGBoost regression models, provided robust and accurate predictions for household expenditures on personal insurance premiums and retirement contributions. Notably, the XGBoost model demonstrated better predictive accuracy compared to the Elastic Net model, highlighting the importance of capturing complex non linear relationships within household financial behaviors. The SHAP analysis further enriched these findings by identifying and quantifying the most influential factors. These insights hold substantial value for policymakers, financial institutions, and businesses looking to tailor economic policies, marketing strategies, and financial products to the nuanced needs and behaviors of different household segments. Future research could expand this analysis by exploring additional demographic and behavioral factors or by applying similar methodologies to other financial contexts and markets, thus enhancing strategic decision making capabilities across various sectors.

Word Count: 3071 words

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